

NC STATE UNIVERSITY

DESIGN OF A CMOS OPERATIONAL TRANSCONDUCTANCE AMPLIFIER USING 90nm TECHNOLOGY

ECE 511 PROJECT REPORT

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Academic Integrity Statement:

I affirm that I have neither given nor received any unauthorized aid during the design and analysis of this project.

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Date: 12-December-2024

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Date: 12-December-2024

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Executive Summary

This report presents the design and analysis process of an Active Cascode Current Mirror Operational Transconductance Amplifier (AC-CM-OTA) using CMOS 90nm technology in Cadence Virtuoso with high gain, wide output swing, high ICMR, CMRR and PSRR. The circuit comprises of a supply independent bias current generation circuit, wide swing cascode optimum bias voltage generation circuit, baby operational amplifier for active cascode, wide swing cascode tail current circuit, and current mirror OTA with wide swing cascode output stage. The input of OTA comprises a NMOS differential input pair with current mirror active load to achieve high transconductance and ensure proper amplification. Wide swing cascode structure was used to boost the gain by increasing the output resistance. Baby Op-Amps were used to boost gain using active cascode technique. An output buffer using a simple source follower topology was implemented to derive output loads as low as $100\ \Omega$ but this was later dropped from the circuit in favor of other specifications. The OTA is biased using a constant g_m biasing circuit with a CMOS inverter as a startup circuit. For voltage biasing of wide swing cascode a single wide swing cascode along with a fixed resistor was used to generate the required voltages. This circuit nominally meets 10 out of 12 specifications and additional specifications of output buffered was dropped in favor of others.

Compliance Table

Table 1: Compliance Table

Parameter	Specification	Achieved
Low Frequency Gain	86 dB (20 kV/V)	68 dB (2511 kV/V)
GBW Product	> 400 MHz	409 MHz
Phase Margin	75°	81°
Output Swing	0.9V <i>V_{pp}</i> single ended	0.93V <i>V_{pp}</i> single ended
Input CM Range	0.8V overlap with output swing	0.93V overlap with output swing
CMRR	> 100dB	108dB
PSRR	> 100dB	93dB
Supply Voltage	1.5V – 1.7V	Nominal Operation Achieved
Power Dissipation	< 4mW	3 mW
Settling Time	< 90ns within 1% of final value	16 ns within 1% of final value
Slew Rate	> 25 V/ μ sec	155 V/ μ sec
Input Referred Noise	12nV/ $\sqrt{\text{Hz}}$	0.004

Output buffer was excluded from the design in favor of other specifications

Roles and Distribution

During the design and development of this project, the tasks have been distributed and completed in an equal manner by both members of the group. Periodic collaboration and exchange of information was done.

Individual contributions

Mian Ahmad Mehmood

- a. OTA Design
- b. Optimum Voltage Generation Circuit Design
- c. Testing and Analysis
- d. Report writing and formatting

Abhishek Ajit Deshpande

- a. Current bias circuit design
- b. Inverter kickstart circuit design
- c. Output buffer design
- d. Testing and Analysis
- e. Report writing and formatting

Technical Discussion

The following sections present a technical discussion of the project. Starting with the design architecture, followed by a short discussion on each of the sub sections of the circuit and their design procedure. In the end a narrative summary of the results is presented.

Block Diagram

The figure gives a signal flow diagram of complete architecture. A bootstrap biasing circuit is used to generate supply independent bias current required for the circuit. This bias current is used for biasing OTA, biasing baby op amps and generating optimum bias voltage for wide swing cascode. Active Cascode Current Mirror OTA topology is used in this project. This has the advantages of high gain and large output swings and GBW. The gain is boosted by active cascode technique and wide output swing is achieved by using wide swing cascode topology. The optimum bias voltage for wide swing cascode is generated using resistor based wide swing cascode circuit to generate a supply independent bias voltage.

System Architecture

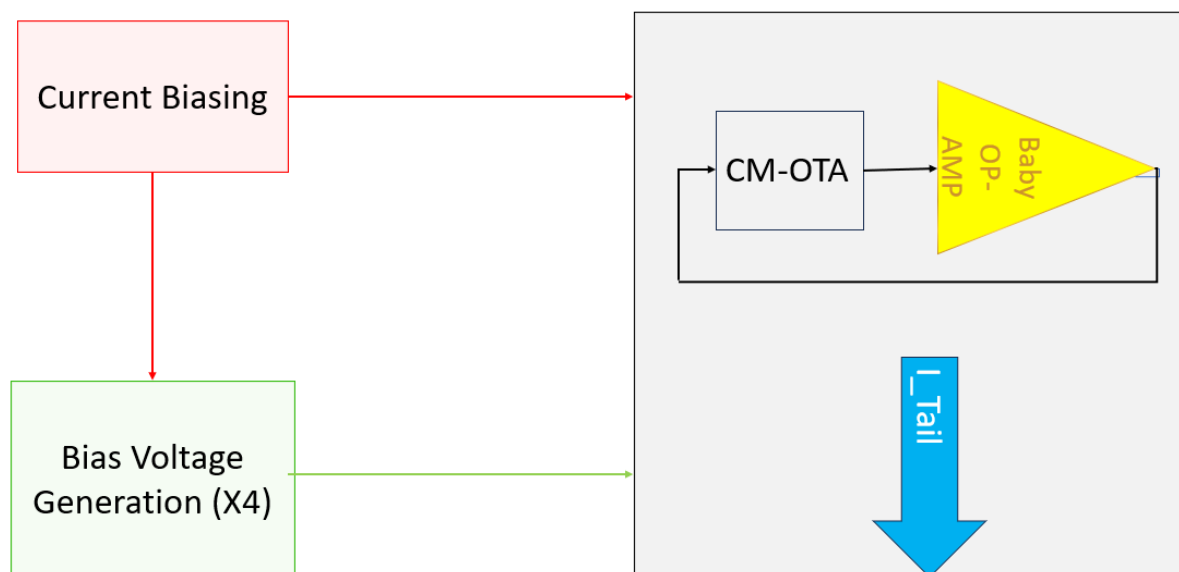


Figure 1: OTA Architecture

Constant gm biasing circuit

The constant g_m biasing topology has been used to create a desired output current of 50uA. It is defined by a stack of PMOS followed by one of NMOS, having diode connections at alternate gates. Bootstrapping is used to create a generated current based on a reference current. There is a R_{set} at the bottom of the output leg which determines the current, along with the sizing of the transistors. The sizing of the NMOS and PMOS was done in a way to keep all the transistors in saturation and give the desired output current.

To jumpstart the circuit from the off state into saturation, a CMOS inverter circuit has been used. In this, the NMOS is kept stronger than the PMOS, to create a pulse that is given as an input to the gate of another NMOS, whose drain connects to the output leg, which in turn starts the flow of current. The NMOS remains in triode and the PMOS goes into saturation, generating the kickstart. Later, a desired bias voltage can be generated using this current or variations in the current by differing mirroring ratios (1: K) to generate desired voltages. These lengths and widths have been adjusted through parametric sweeps to get proper startup of the biasing circuit. For proper current generation as per empirical results,

$$K = 1.93$$

$$R_{set} = 1.5 \text{ k}\Omega$$

Active Cascode Current Mirror OTA

For the given specifications Current Mirror OTA topology was selected to start with. This topology has good gain, which can be further boosted using active cascading technique. It has good input and output voltage swings which were required for the design. Due to it being a single stage it also has good gain bandwidth products which can be managed easily compared to two stage design.

For the design considering power consumption, 60% of the power consumption was dedicated to the OTA and 40% was dedicated to the rest of the circuit.

$$I = \frac{P}{V} = \frac{4 * 10^{-3}}{1.7} = 2.3 \text{ mA}$$

$$I_{OTA} = 0.6 * 2.3 = 1.4 \text{ mA}$$

This current was again divided into the current for OTA and Active Cascode, which yields about $I_{OTA} = 1 \text{ mA}$ and $I_{AC} = 0.4 \text{ mA}$. Therefore, each branch was kept at a maximum of $250 \mu\text{A}$ to meet the power consumption requirements. Another important factor for the selection of bias current was the required W_{byL} to achieve a $V_{eff} = 0.1 \text{ V}$.

This requirement is dictated by the output swing requirement. Assuming a threshold voltage of 250 mV

$$V_{bn} - V_{th} < V_O < V_{bp} + V_{th}$$

$$0.5 - 0.25 < V_O < 1.0 + 0.25$$

$$0.25 \text{ V} < V_O < 1.25$$

This gives an output swing of 1 V . For these given requirements W_{byL} needed to be large and I_D needed to be around $100 \mu\text{A}$ which results in a $V_{eff} = 0.1 \text{ V}$.

Table 2: W_{byL} and V_{eff} Calculations with Bias Current

Id (uA)	0.1	0.2	0.3	0.4	0.5
Veff (V)	WbyL	WbyL	WbyL	WbyL	WbyL
0.1	200	400	600	800	1000
0.2	50	100	150	200	250
0.3	22.22222	44.44444	66.66667	88.88889	111.1111
0.4	12.5	25	37.5	50	62.5
0.5	8	16	24	32	40

To get a wide output swing the bias voltage of the active cascode needs to be low which is controlled by

$$V_{gs} = V_{th} + V_{eff} = 0.25 + 0.1 = 0.35 \text{ V}$$

These values serve as a starting point for the design which would meet the power consumption and output swing requirements. To get a wide input common mode voltage, the V_{gs} of the tail transistors needed to be in the same range as above. To achieve this wide swing cascode technique used for giving the bias current of the circuit.

The gain requirement was divided into three sections to facilitate a smooth design procedure. In stage 1 a simple Current Mirror OTA would be designed to have a gain of 30dB. In stage 2 the output of this simple OTA would be cascode to produce high output impedance and give a total gain of 65dB. In the third stage active cascode would be added to further boost the gain by 25 dB to get to the final value of 86 dB gain.

A compensation capacitor is added at the output of the OTA to act as a dominant pole and give a fixed gain bandwidth product. This capacitor was accompanied by a resistor to set the zero for adding 30° phase margin to get to the final specification value of 75°. The compensation capacitor was calculated to push the dominant pole to the maximum limit to get the highest possible GBW.

Wide Swing Cascode Optimum Bias Voltage Generation

To generate optimum bias voltage a resistor based wide swing cascode circuit was implemented. The value of the resistor was then swept to find the optimum value for each wide swing cascode. This topology has the advantage of being simple and generating supply independent voltage bias for wide swing cascode, but it is only suitable for a fixed current.

Output Buffer

For this project, the simple source follower was designed to be used as an output buffer. The dimensions of the NMOS used are as shown in the figure above. It receives the output from the OTA and delivers it to the load. However, in this case, for design simplicity, the buffer has been created in a no-load condition, to reduce the fluctuations in the parameters. The source resistor value was passed through a DC sweep to find out the value of R_s such that the reduction in gain and other critical parameters is reduced. In favor of other specifications, output buffer was dropped from the design.

Narrative Summary

For the design, a gain of 68dB was achieved with a GBW of 409 MHz and Phase Margin of 81°. The output swing of 0.93V was achieved, which had an overlap of 0.93V with the input common mode voltage range. 108dB of CMRR was achieved and 93dB of PSRR was achieved due to a lower gain of 68dB. A compromise between output swing and gain was made to get good output swing at the cost of lower gain. Settling time of 16ns was achieved and a slew rate of 154 V/us was achieved indicating excellent transient performance of the circuit. Input referred noise specification was not met.

Schematic set 1

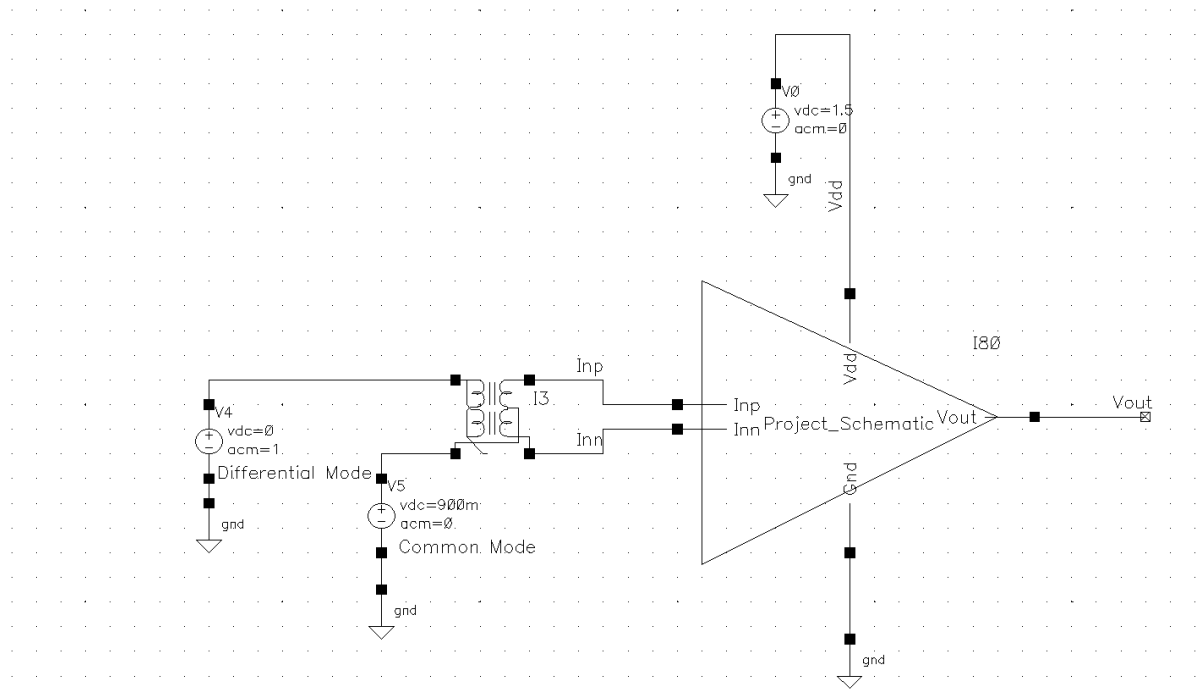


Figure 2: Project Test Bench Schematic

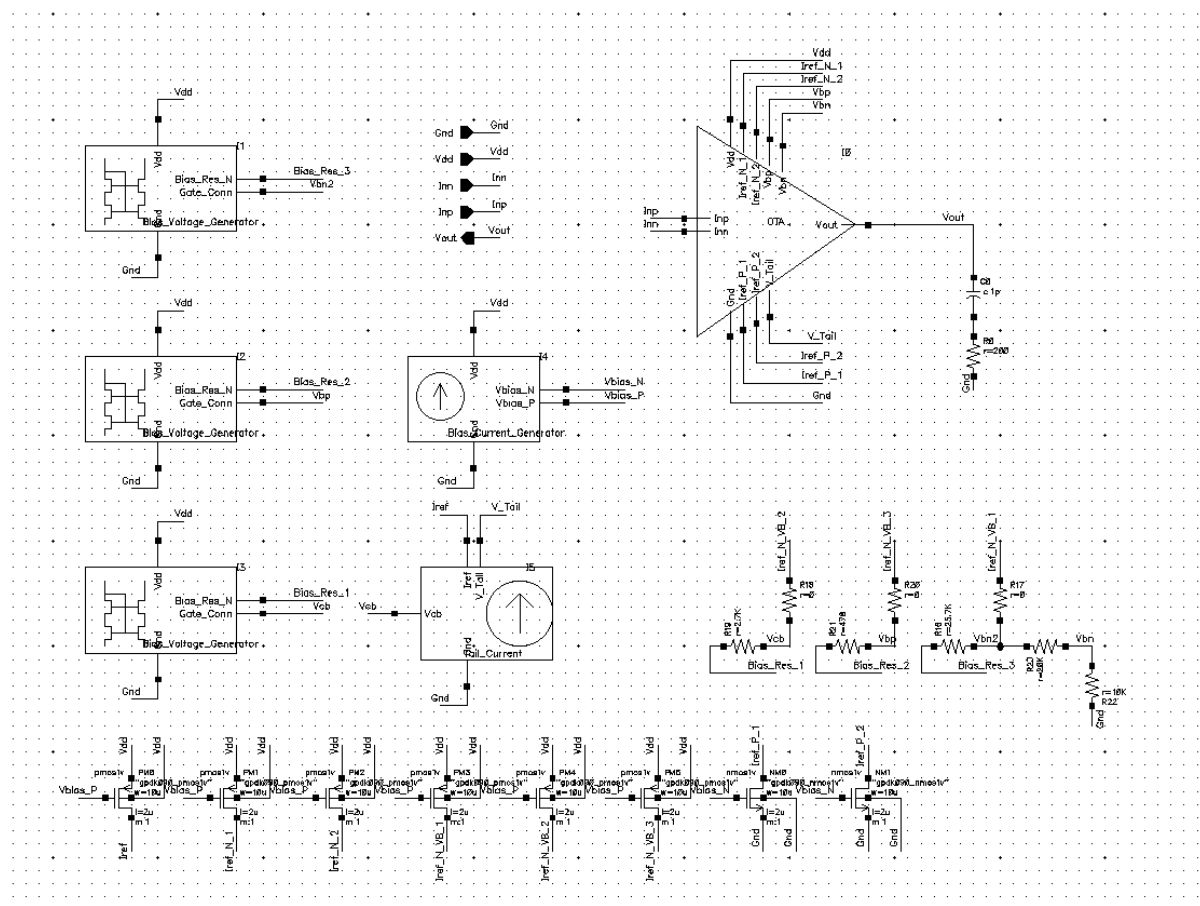
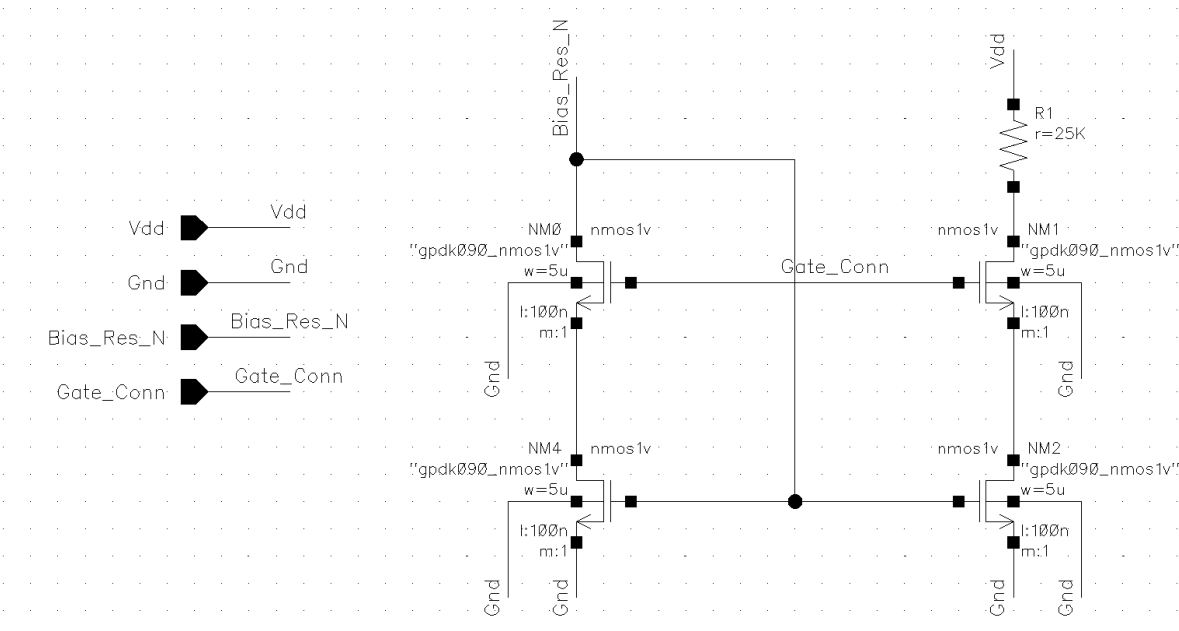
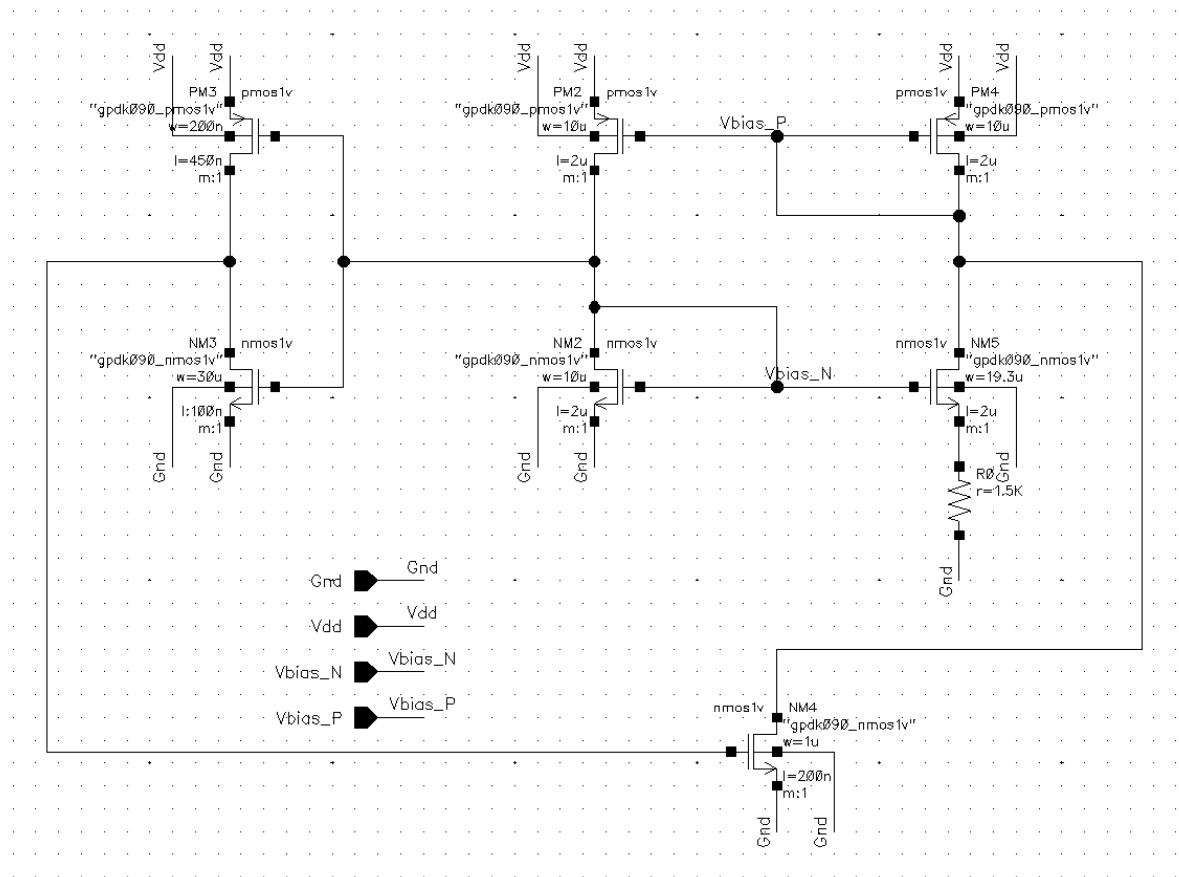


Figure 3: Complete project Schematic



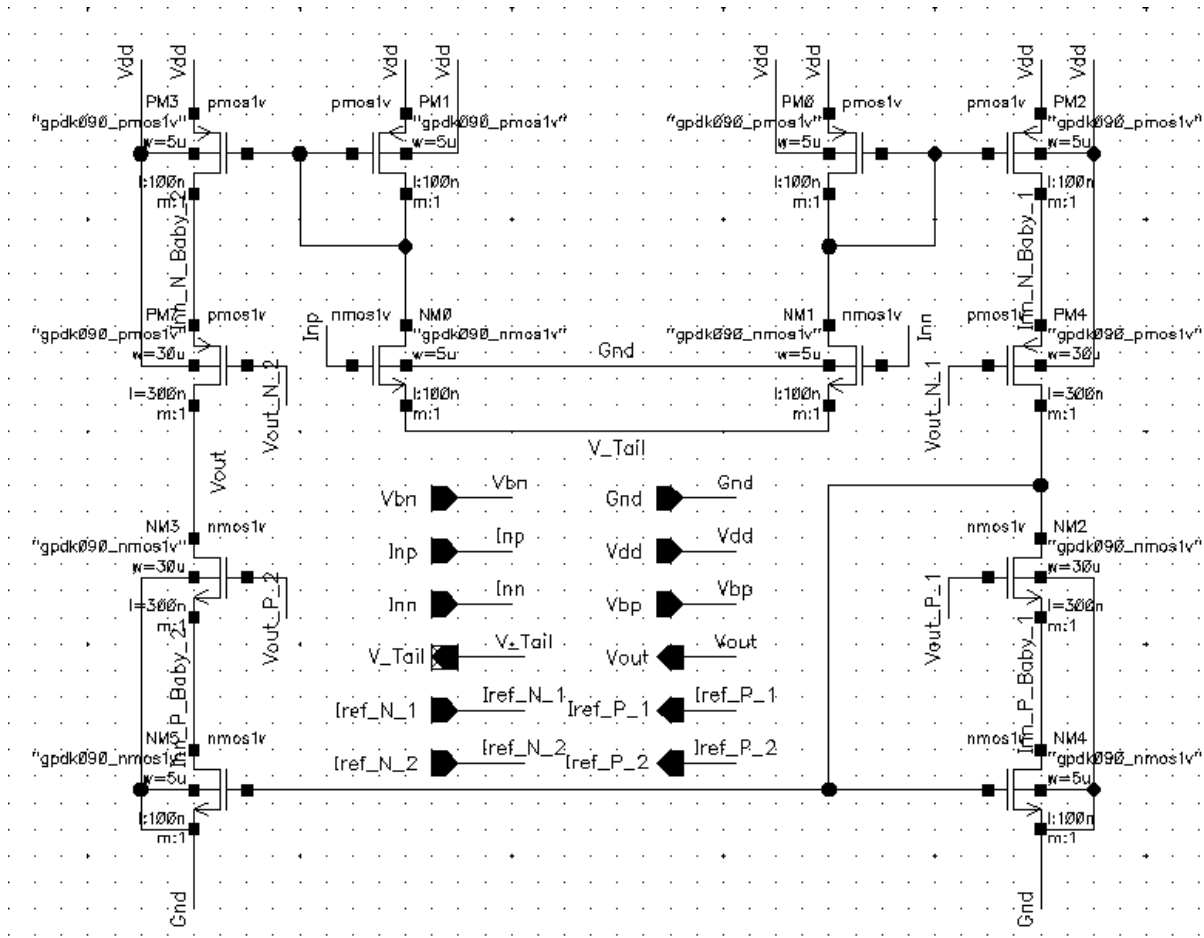


Figure 8: Current Mirror OTA Circuit

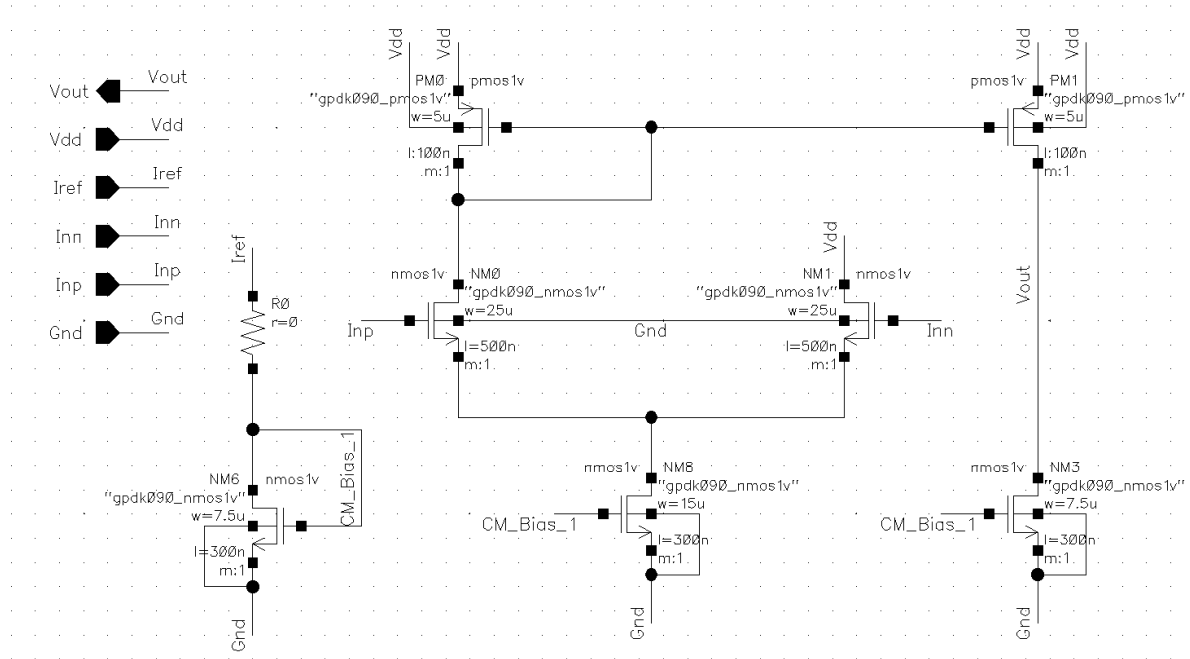


Figure 9: NMOS Baby Op Amp

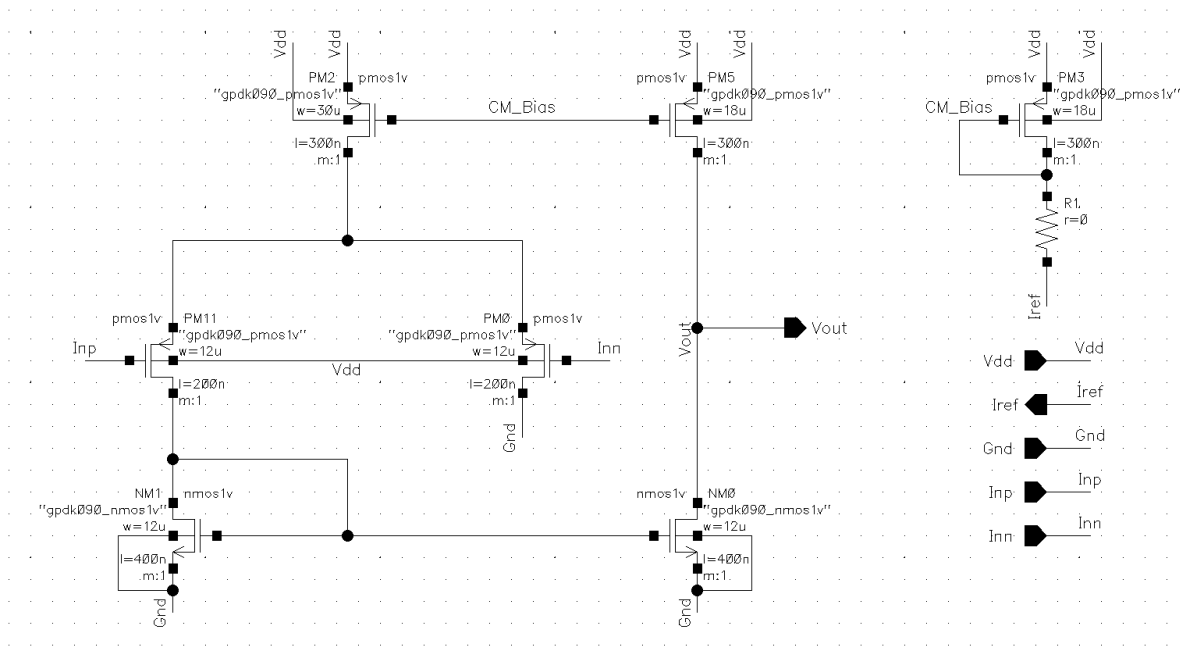


Figure 10: PMOS Baby Op Amp

Results

Adding an output buffer to derive low impedance loads, deteriorates the achieved results. For this reason the results are discussed for both cases, with and with out output buffer.

Results with Output Buffer

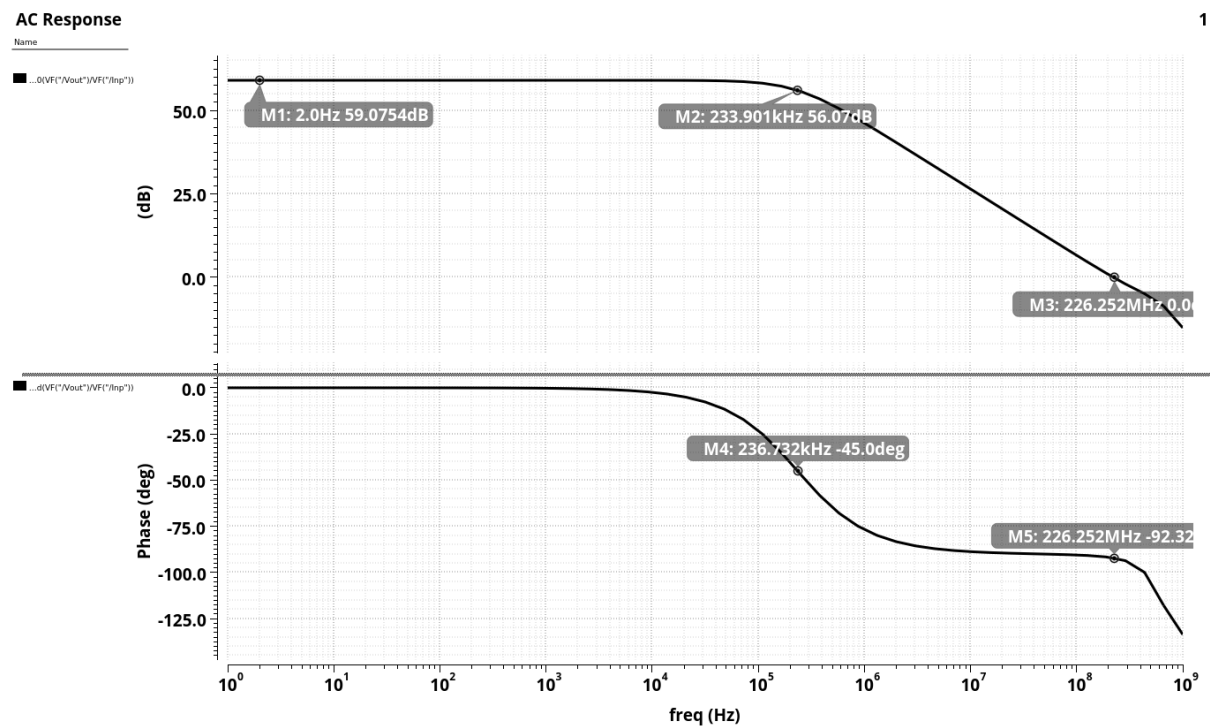


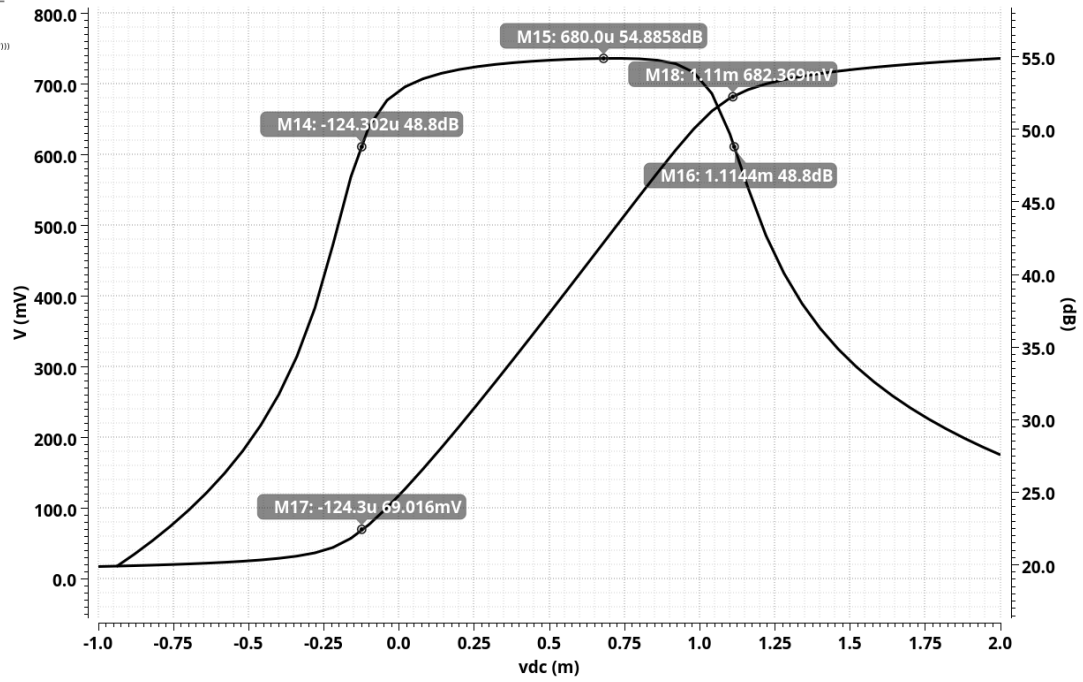
Figure 11: Gain, Phase margin and GBW

DC Response

Tue Dec 3 11:35:29 2024 1

Name

■ Vout
 ■ (v("Nout" ?result "dc"))

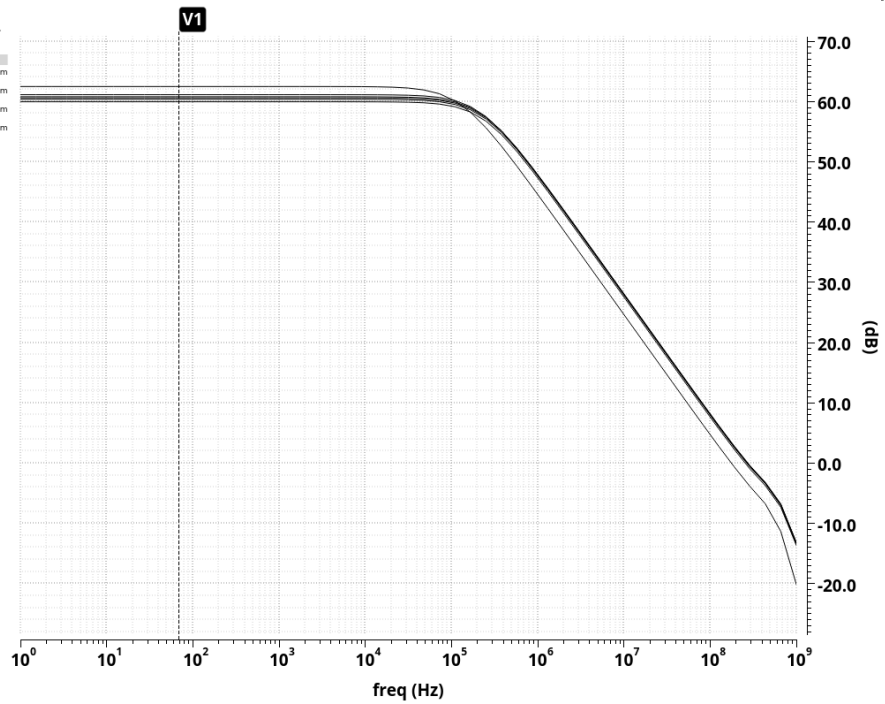


AC Response

1

Name V1 Vicm

#B20(VF1("Nout")/VF1("Inp"))	62.4804dB	300.0m
#B20(VF1("Nout")/VF1("Inp"))	61.1231dB	480.0m
#B20(VF1("Nout")/VF1("Inp"))	60.7874dB	660.0m
#B20(VF1("Nout")/VF1("Inp"))	60.5877dB	840.0m
#B20(VF1("Nout")/VF1("Inp"))	60.3419dB	1.02
#B20(VF1("Nout")/VF1("Inp"))	59.9503dB	1.2



AC Response

1

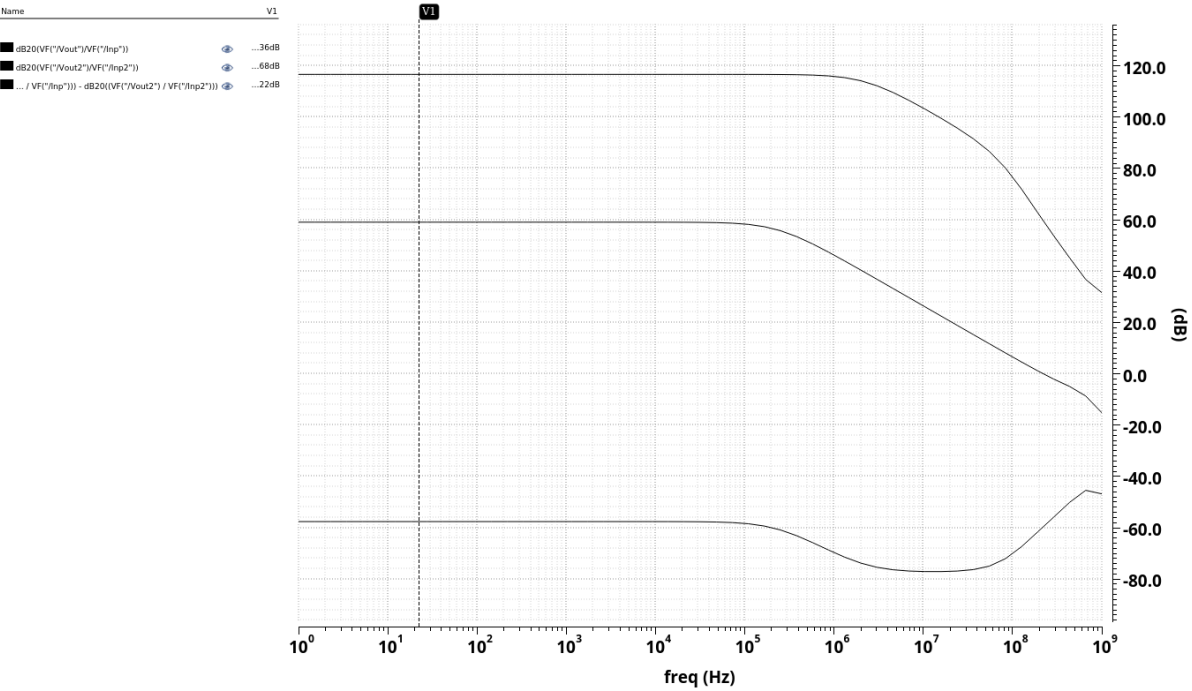


Figure 14: CMRR

AC Response

1

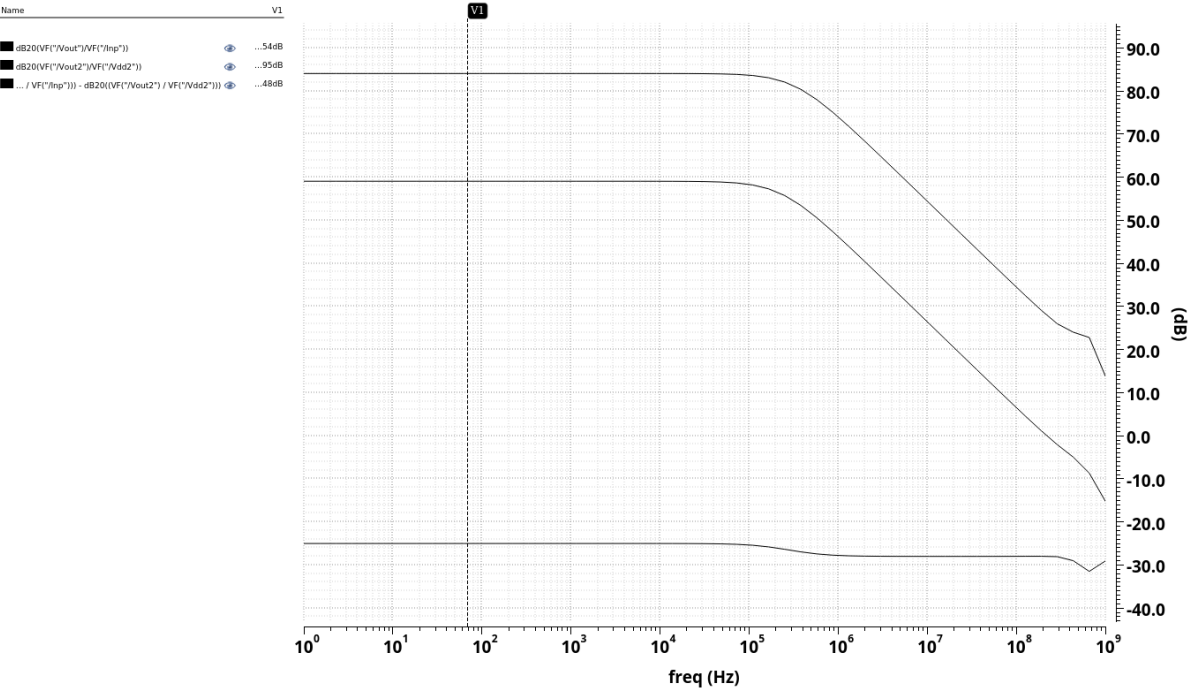


Figure 15: PSRR

Results without Output Buffer – Vdd =1.5V

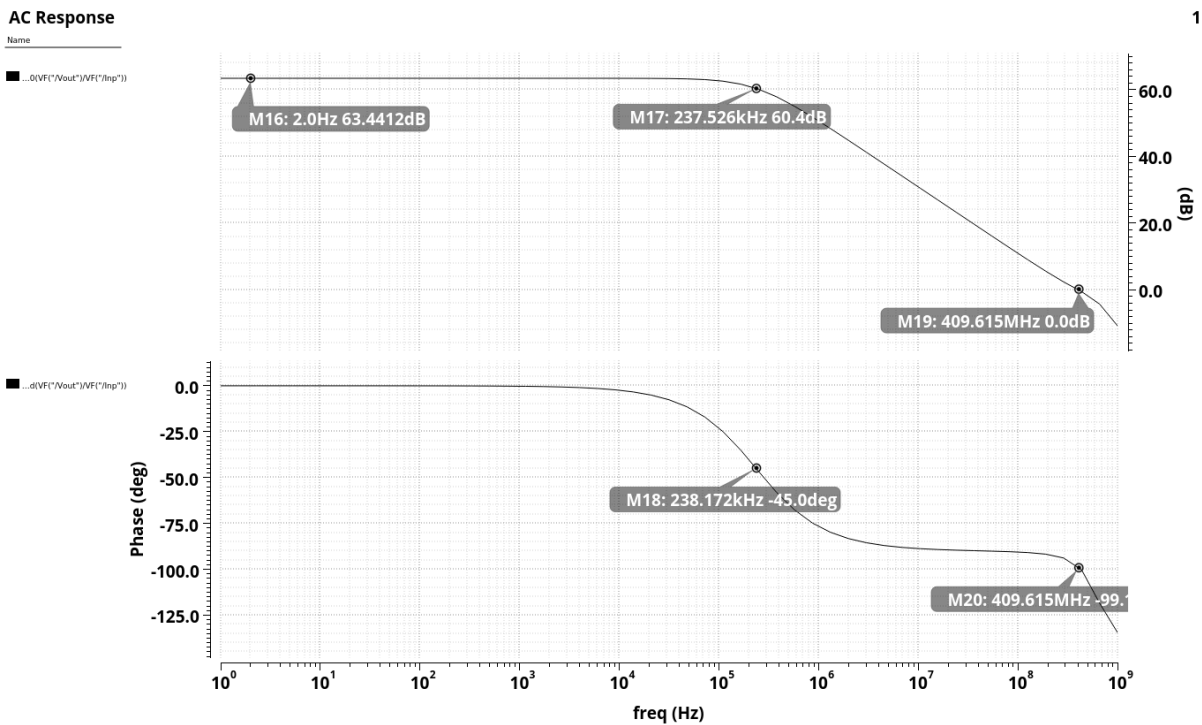


Figure 16: Gain, Phase margin and GBW

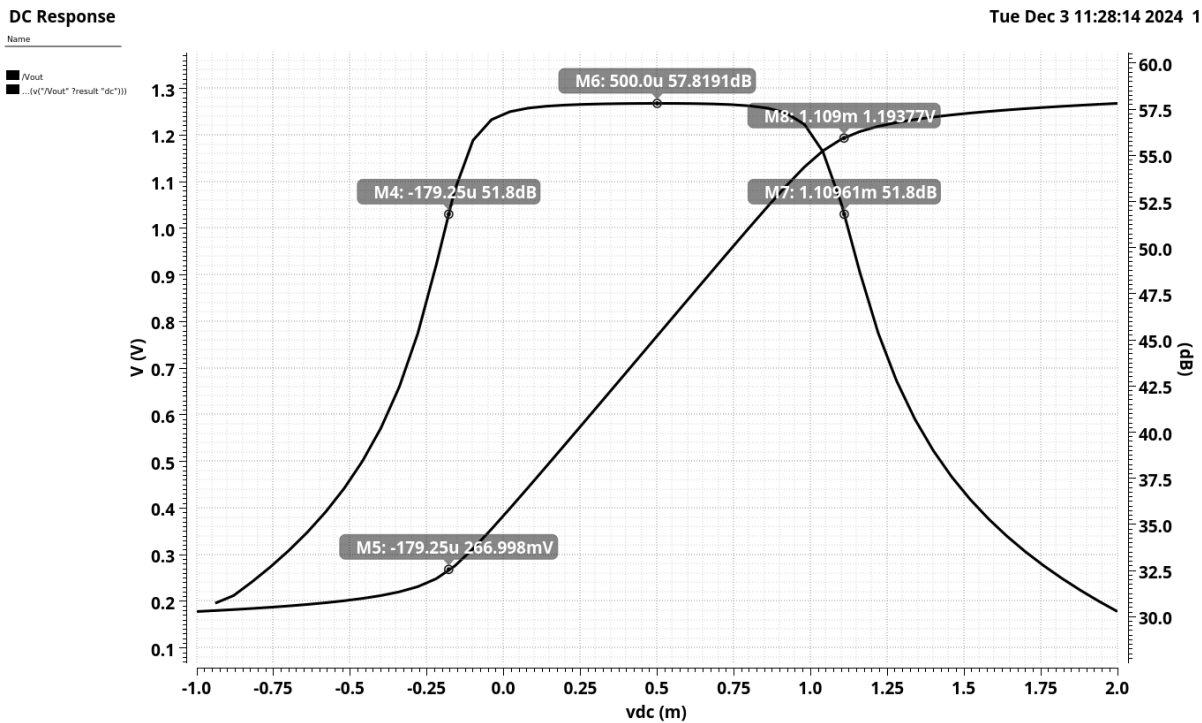


Figure 17: Output Swing

AC Response

1

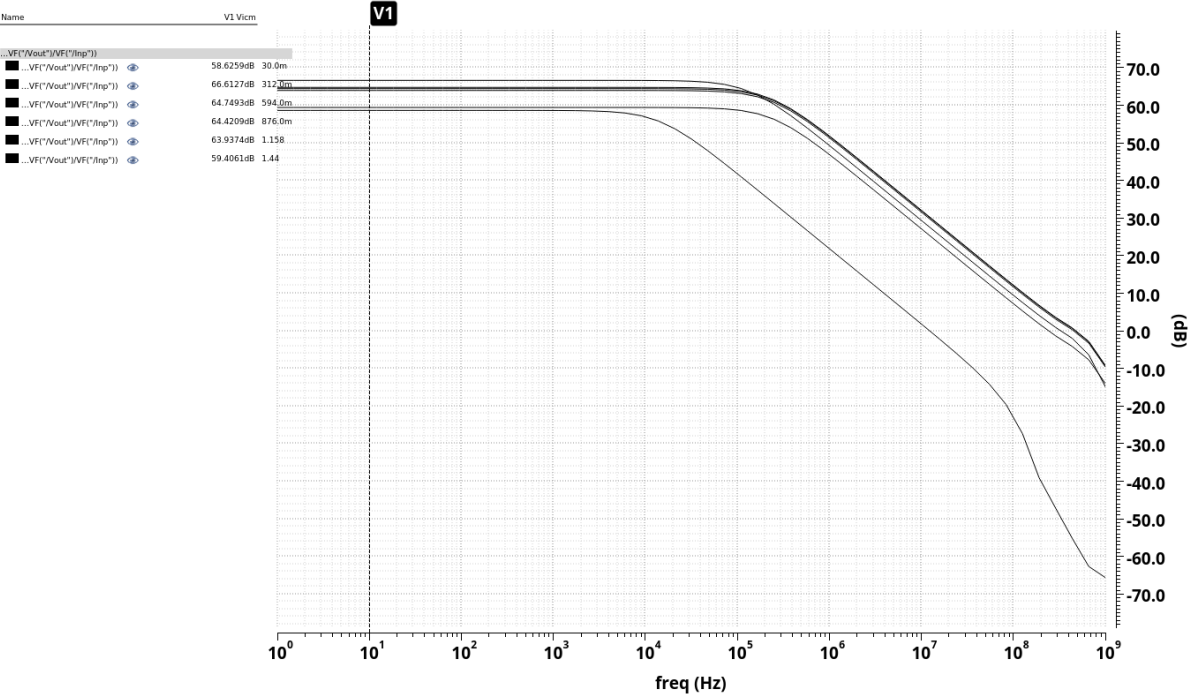


Figure 18: ICMR

AC Response

1

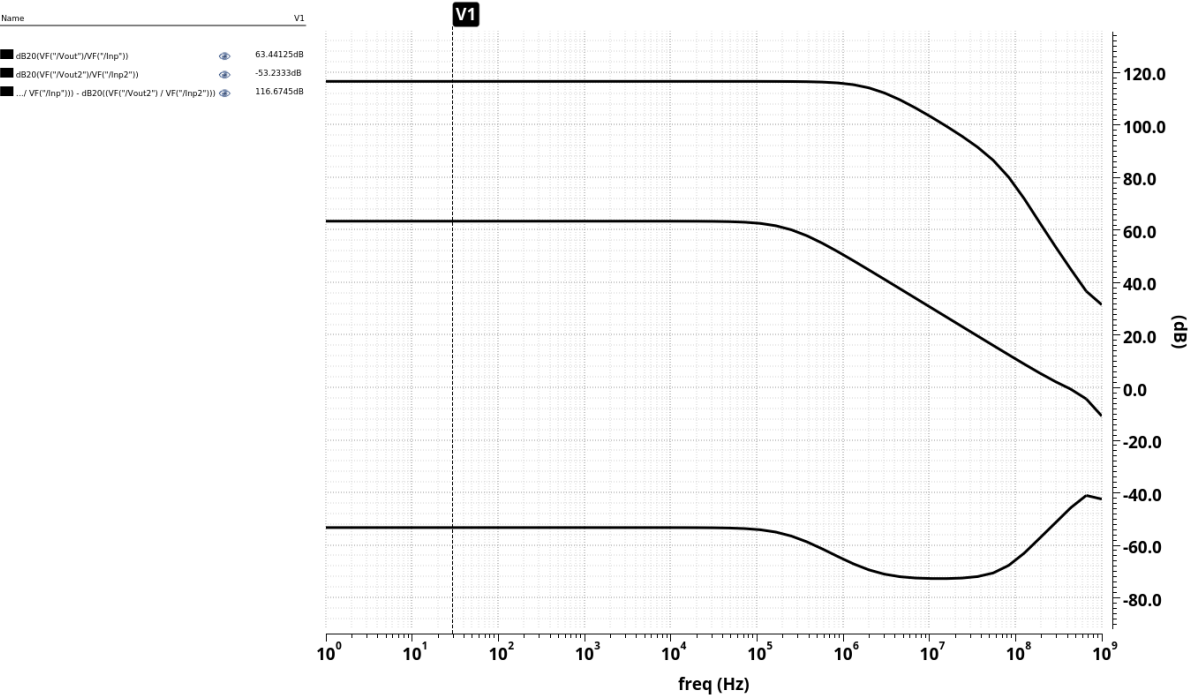
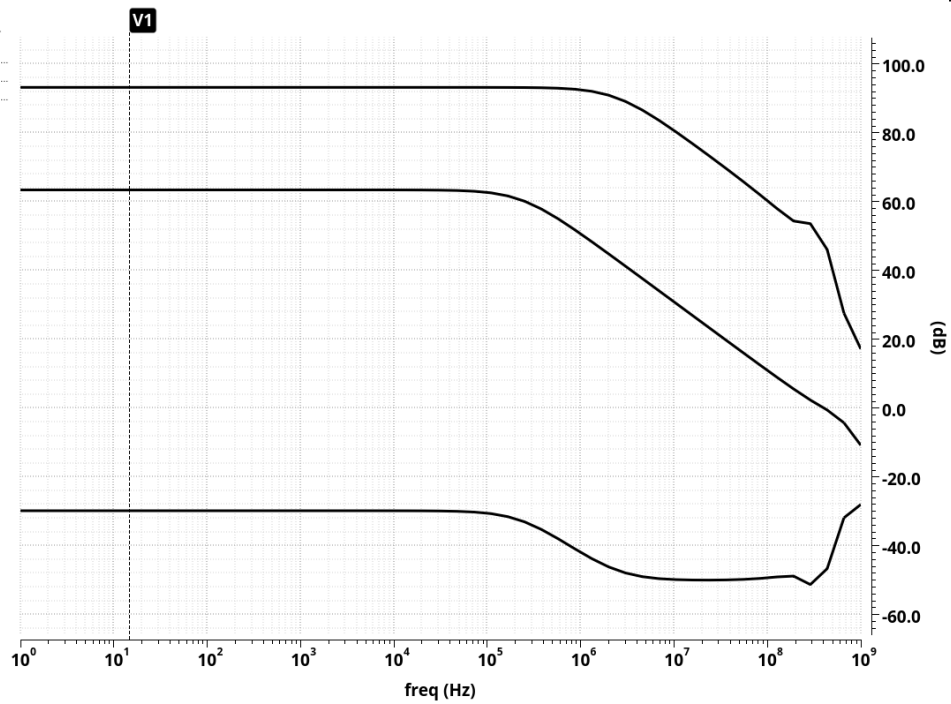


Figure 19: CMRR

AC Response

Name		
dB20(VF1("Vout1")/VF1("Iinp"))	...	...
dB20(VF1("Vout2")/VF1("Vdd2"))	...	...
F("Iinp1") - dB20(VF1("Vout2") / VF1("Vdd2"))	...	...



signal	OP("/V0" "??")
i	-1.93961m
pwr	-2.90941m
v	1.5

Figure 21: Power Dissipation

Transient Response

Tue Dec 3 15:05:41 2024 1

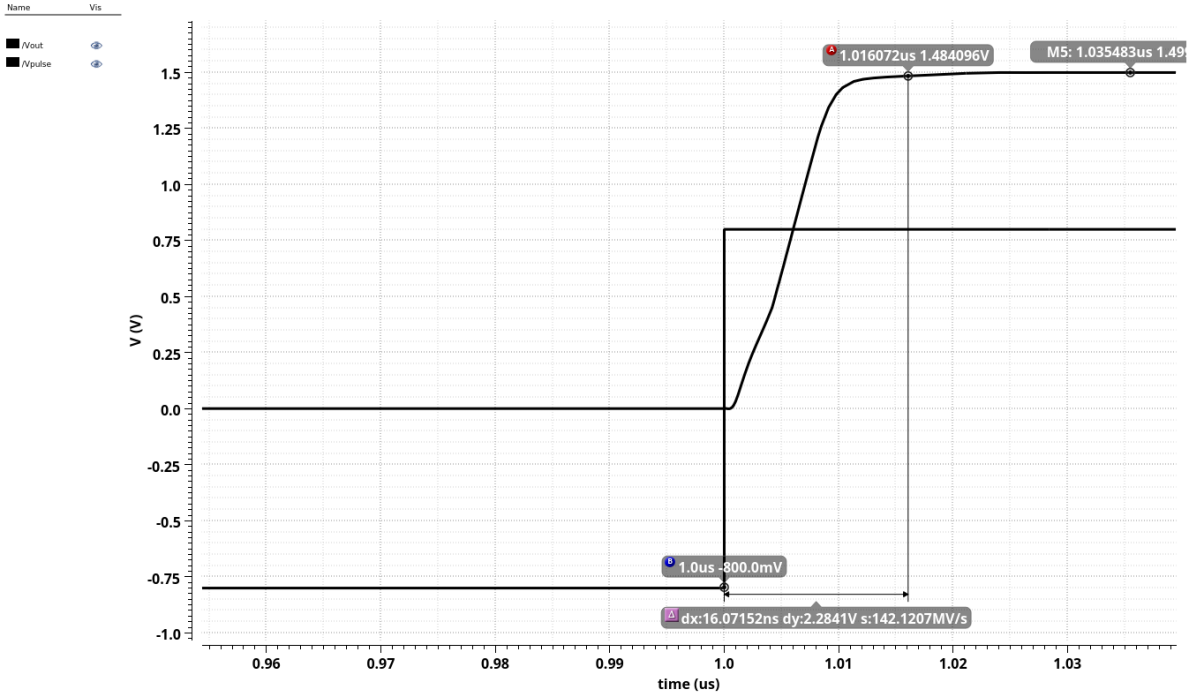


Figure 22: Settling Time

Transient Response

Tue Dec 3 15:12:40 2024 1

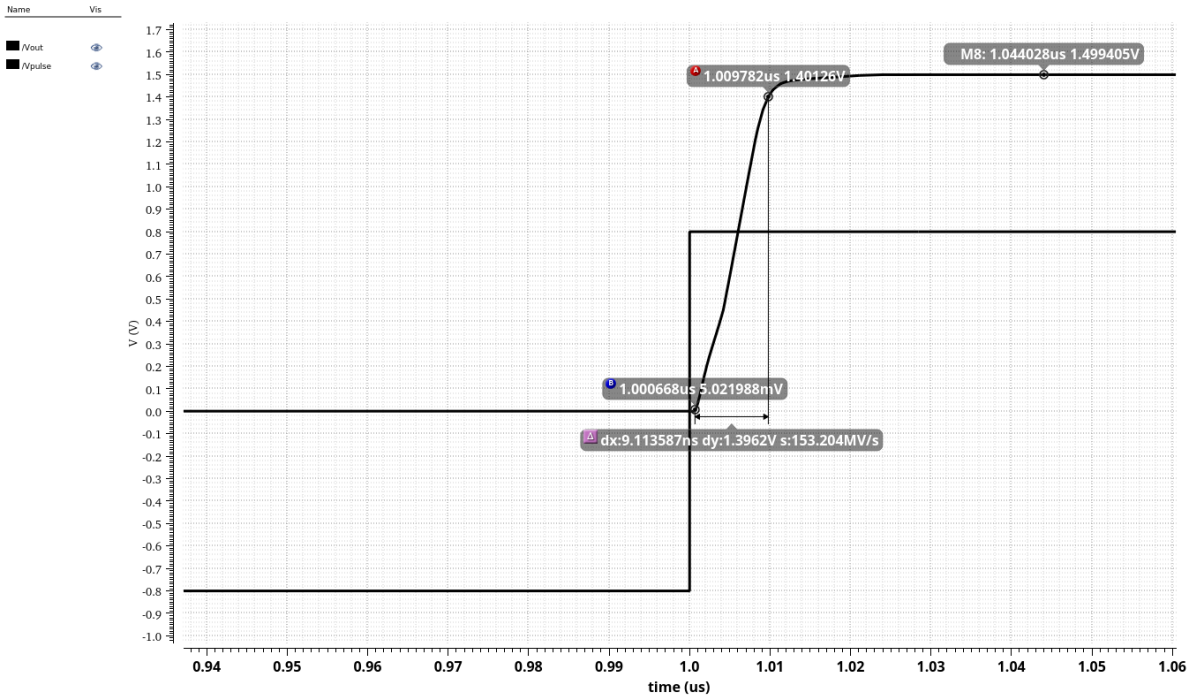


Figure 23: Slew Rate

Noise Response

Tue Dec 3 14:03:12 2024 1

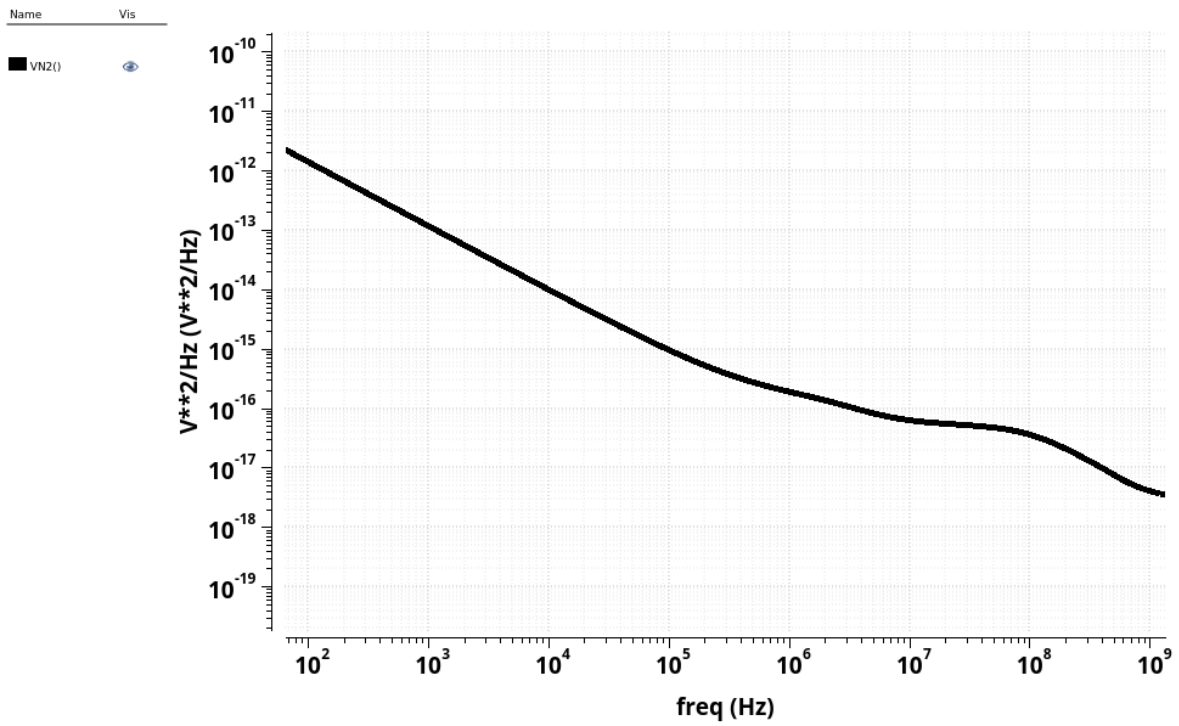


Figure 24: Input Referred Noise

Device	Param	Noise Contribution	% Of Total
/I77/PM3	fn	1.24555e-05	32.77
/I77/PM1	fn	1.23136e-05	32.03
/I77/NM1	fn	6.33422e-06	8.48
/I77/NM0	fn	4.92147e-06	5.12
/I77/NM6	fn	4.76153e-06	4.79
/I77/PM2	fn	4.70091e-06	4.67
/I77/NM5	fn	3.99316e-06	3.37
/I77/PM0	fn	2.78131e-06	1.63
/I77/NM8	fn	2.74512e-06	1.59
/I81/PM4	fn	1.86779e-06	0.74
/I81/PM4	id	1.47149e-06	0.46
/I81/PM2	fn	1.44883e-06	0.44
/I81/NM2	id	1.33194e-06	0.37
/I77/NM5	id	1.24952e-06	0.33
/I81/NM5	id	1.13472e-06	0.27
/I81/PM2	id	1.1315e-06	0.27
/I77/NM9	fn	1.11729e-06	0.26
/I77/PM3	id	1.10133e-06	0.26
/I81/R0	rn	1.03325e-06	0.23
/I77/I7/NM3	fn	1.02875e-06	0.22
Integrated Noise Summary (in V) Sorted By Noise Contributors			
Total Summarized Noise = 2.17575e-05			
Total Input Referred Noise = 4.4163e-05			
The above noise summary info is for noise data			

Figure 25: Input referred noise value

Results without Output Buffer – Vdd = 1.7V

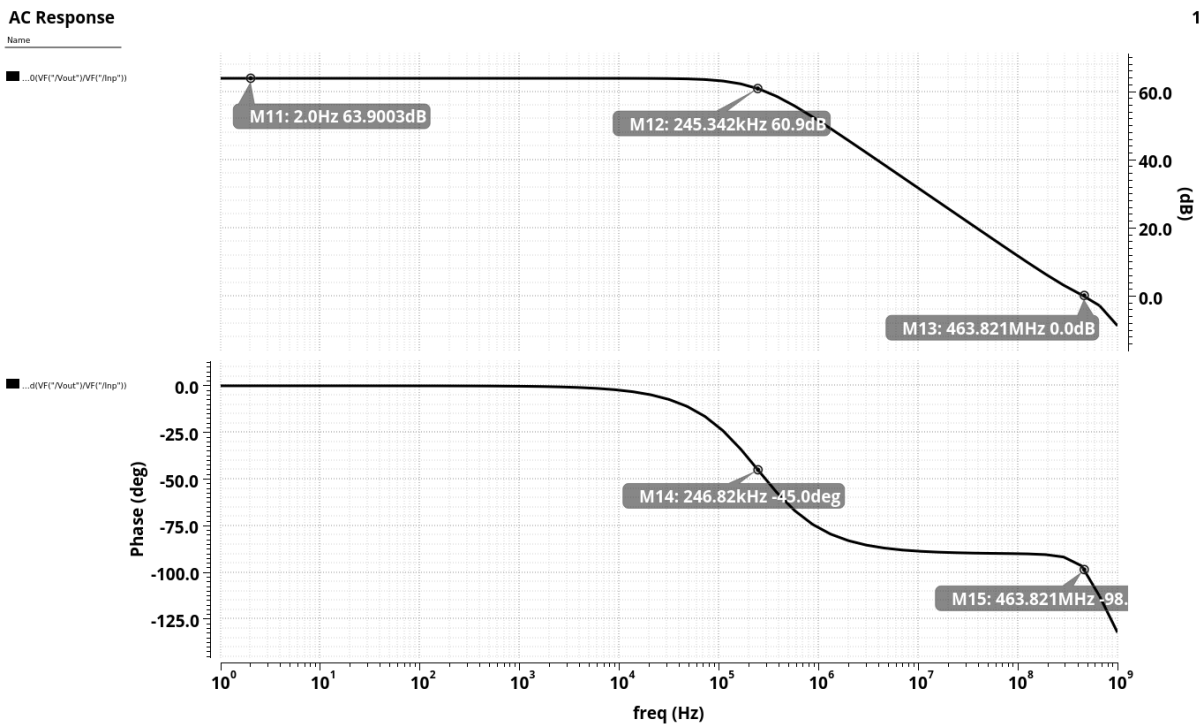


Figure 26: Gain, Phase Margin and GBW

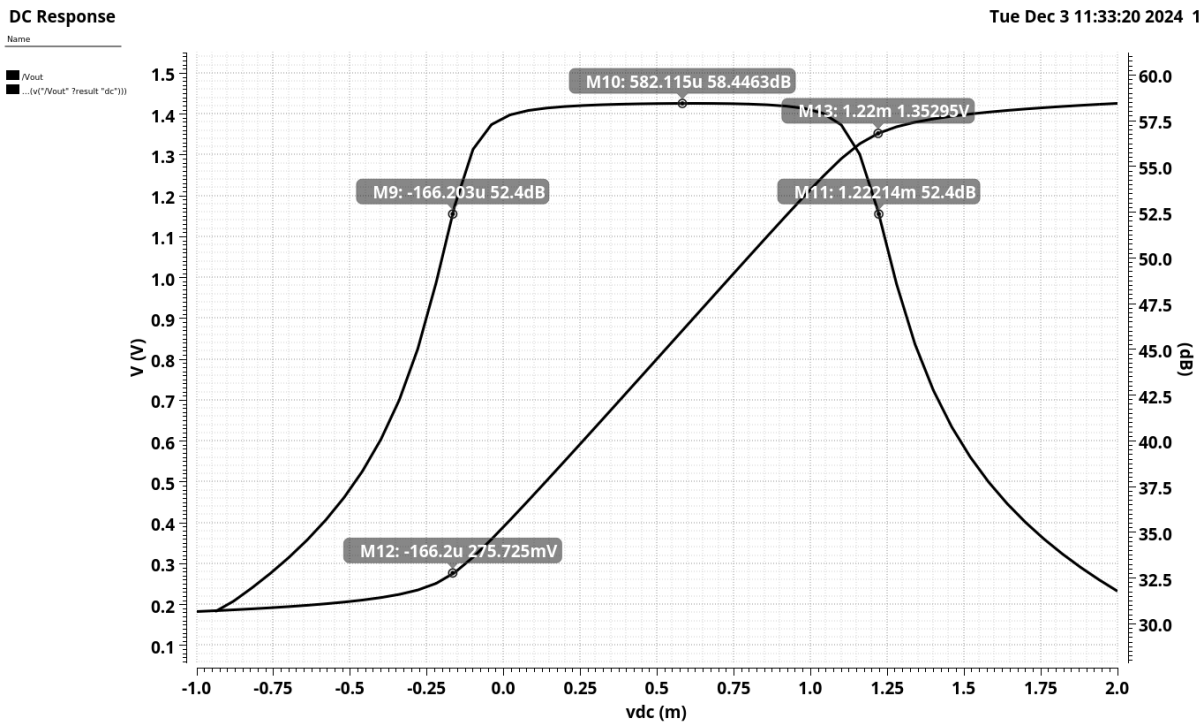


Figure 27: Output Swing

AC Response

1

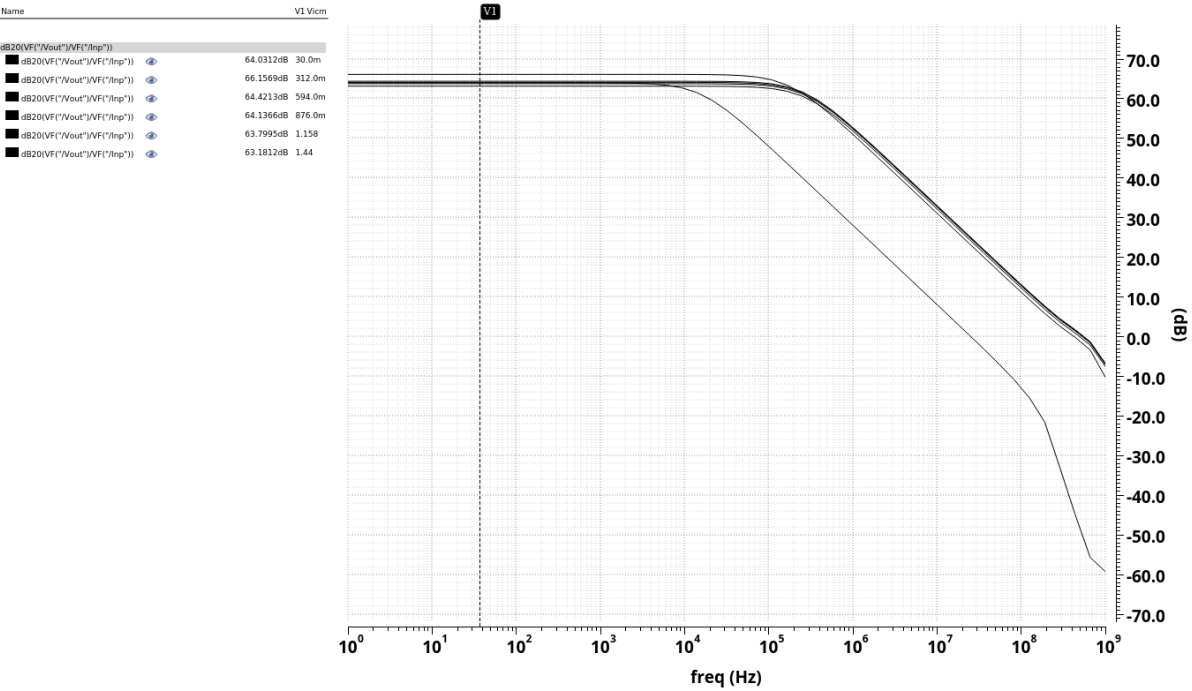


Figure 28: ICMR

AC Response

1

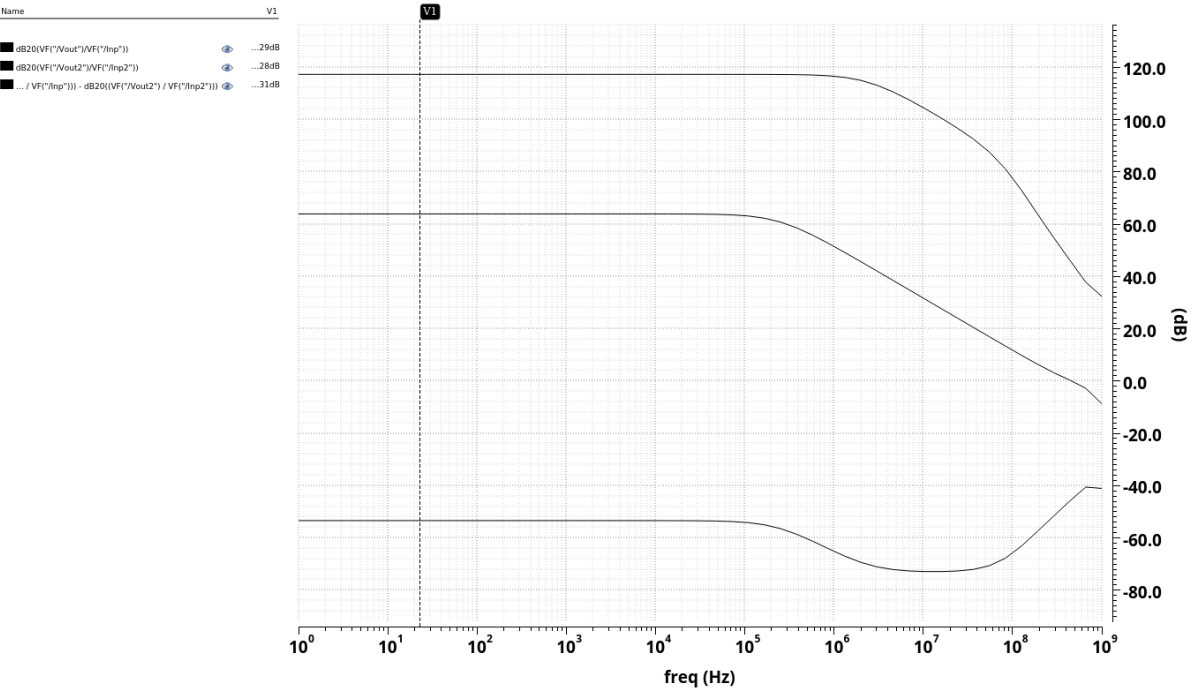


Figure 29: CMRR

AC Response

Name		
dB20(VF1("Vout1")/VF1("Iinp"))	...	...
dB20(VF1("Vout2")/VF1("Vdd2"))	...	...
F("Iinp1") - dB20(VF1("Vout2") / VF1("Vdd2"))	...	...

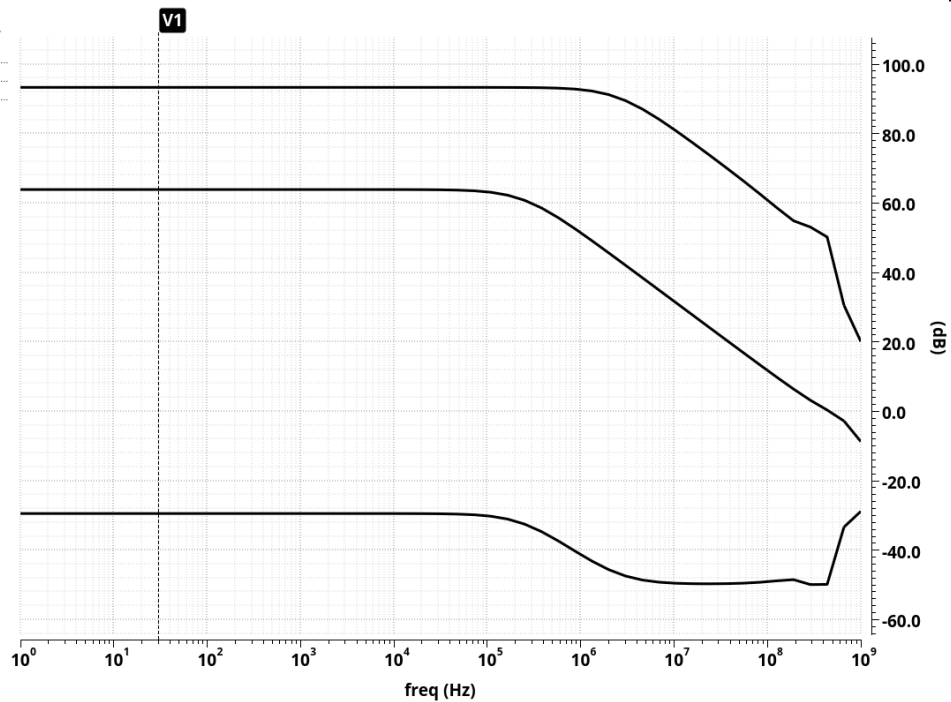


Figure 30: PSRR

signal	OP("/V0" "??")
i	-2.13849m
pwr	-3.63543m
v	1.7

Figure 31: power dissipation

Transient Response

Tue Dec 3 15:09:30 2024 1

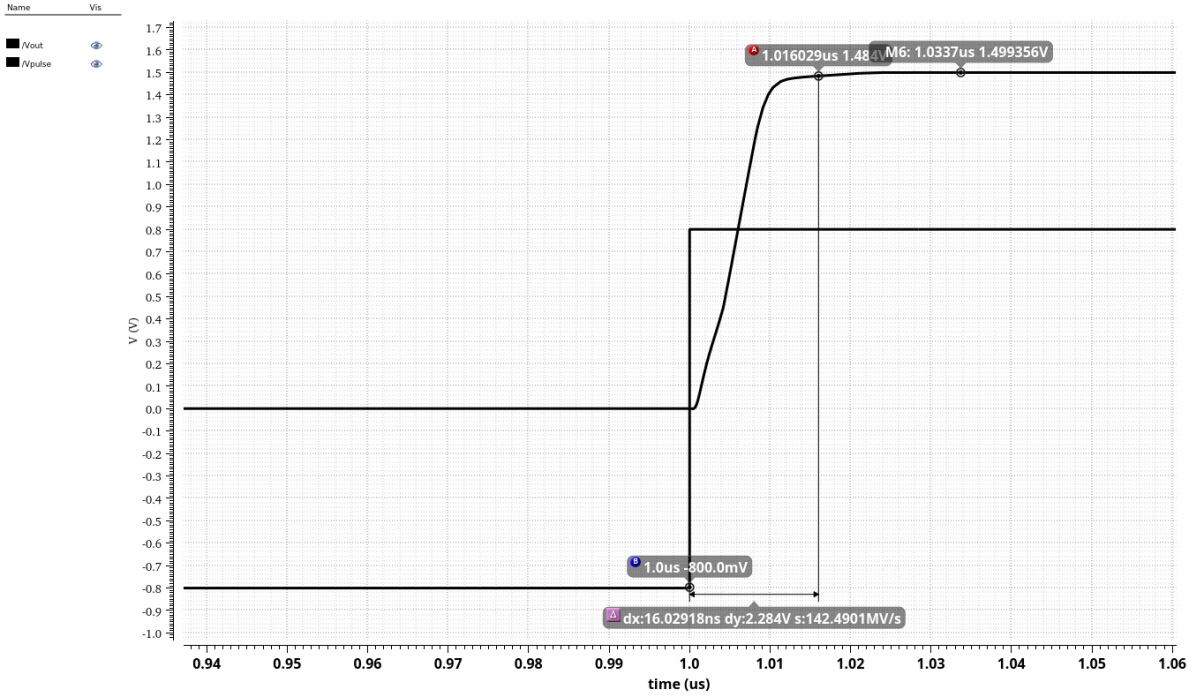


Figure 32: Settling Time

Transient Response

Tue Dec 3 15:09:30 2024 1

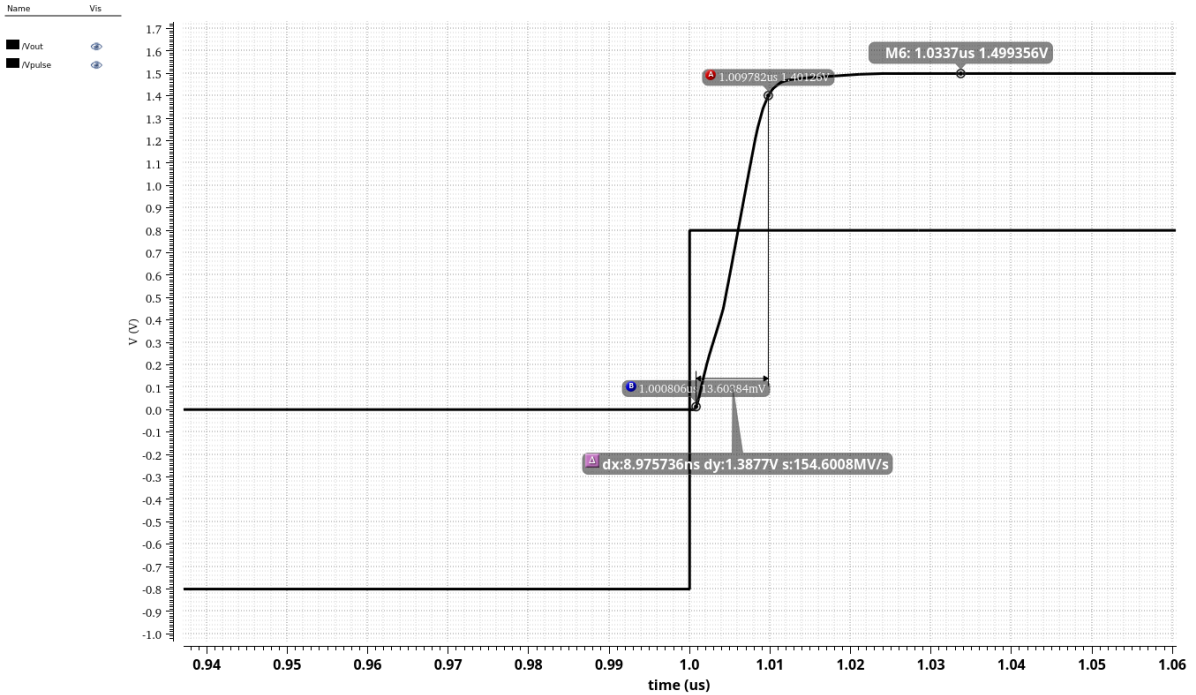


Figure 33: Slew Rate

Noise Response

Tue Dec 3 14:10:24 2024 1

Name Vis

■ VN2()

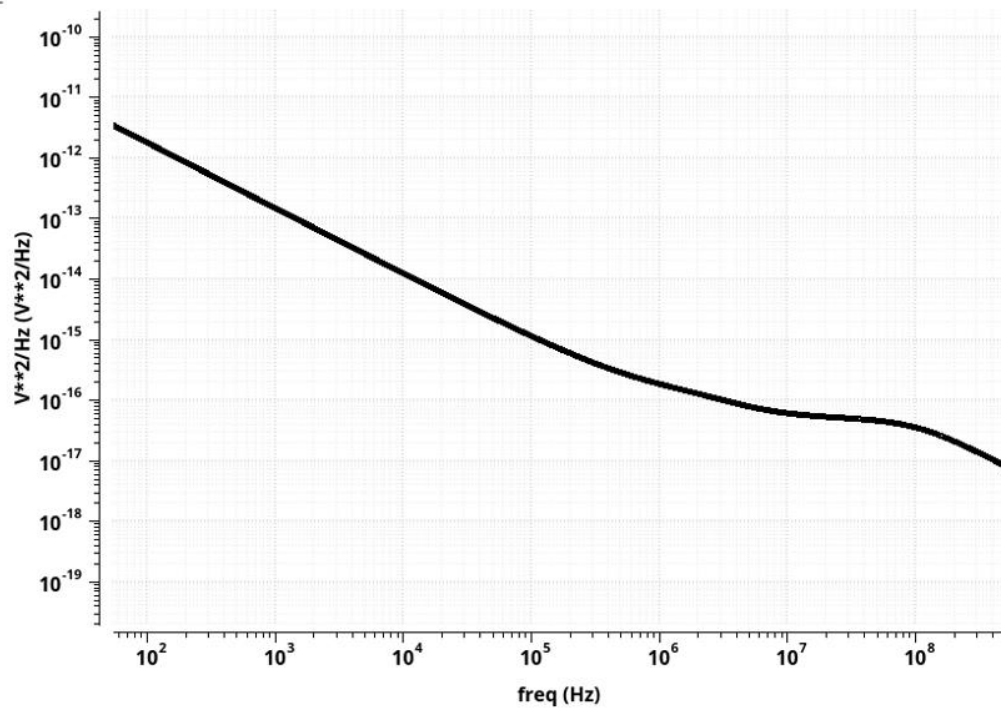


Figure 34: Input Referred Noise

Device	Param	Noise Contribution Total	% Of
/I77/PM3	fn	1.56183e-05	41.65
/I77/PM1	fn	1.27015e-05	27.54
/I77/NM1	fn	6.93153e-06	8.20
/I77/NM0	fn	5.11446e-06	4.47
/I77/PM2	fn	5.03544e-06	4.33
/I77/NM6	fn	4.94872e-06	4.18
/I77/NM5	fn	4.16047e-06	2.96
/I77/PM0	fn	3.02866e-06	1.57
/I77/NM8	fn	2.72106e-06	1.26
/I81/PM4	fn	1.61149e-06	0.44
/I81/PM2	fn	1.24555e-06	0.26
/I81/PM4	id	1.23734e-06	0.26
/I77/NM5	id	1.20152e-06	0.25
/I81/NM2	id	1.11476e-06	0.21
/I77/NM9	fn	1.10781e-06	0.21
/I77/PM3	id	1.0541e-06	0.19
/I77/I7/NM3	fn	1.04786e-06	0.19
/I81/PM2	id	9.45793e-07	0.15
/I81/NM5	id	9.44452e-07	0.15
/I81/R0	rn	8.72251e-07	0.13
Integrated Noise Summary (in V) Sorted By Noise Contributors			
Total Summarized Noise = 2.42012e-05			
Total Input Referred Noise = 4.72014e-05			
The above noise summary info is for noise data			

Figure 35: Input referred noise values

Appendix – Schematic Set 2

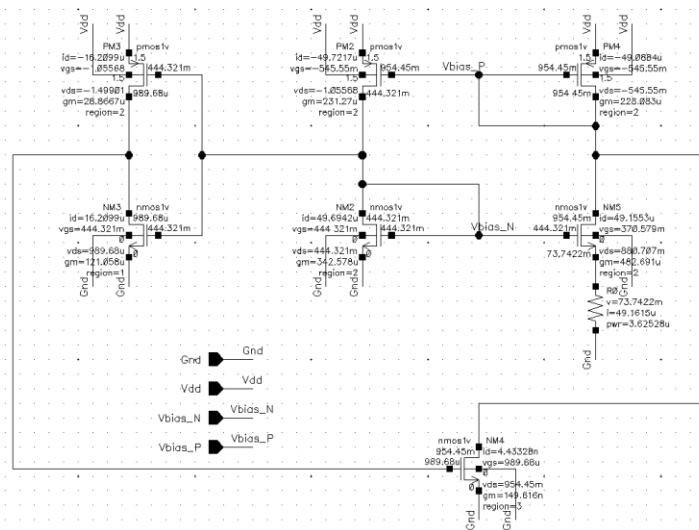


Figure 36: Current Biasing Circuit

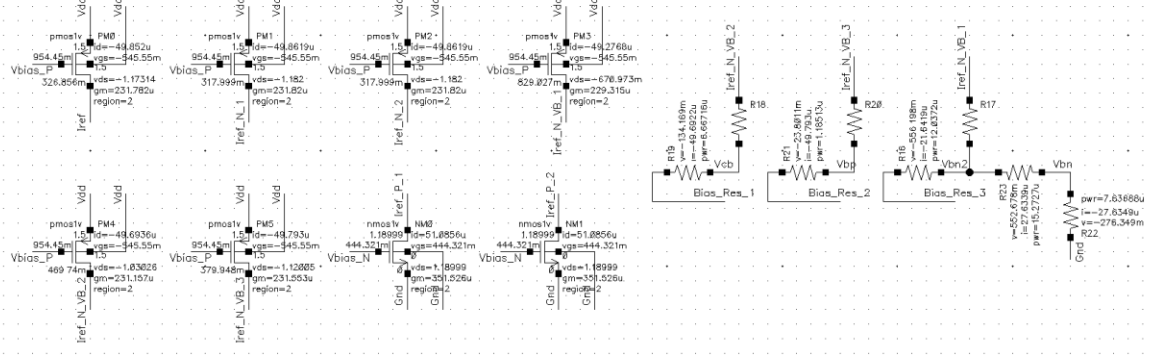


Figure 37: Current Mirror Legs and Wide Swing Voltage Biasing Circuit

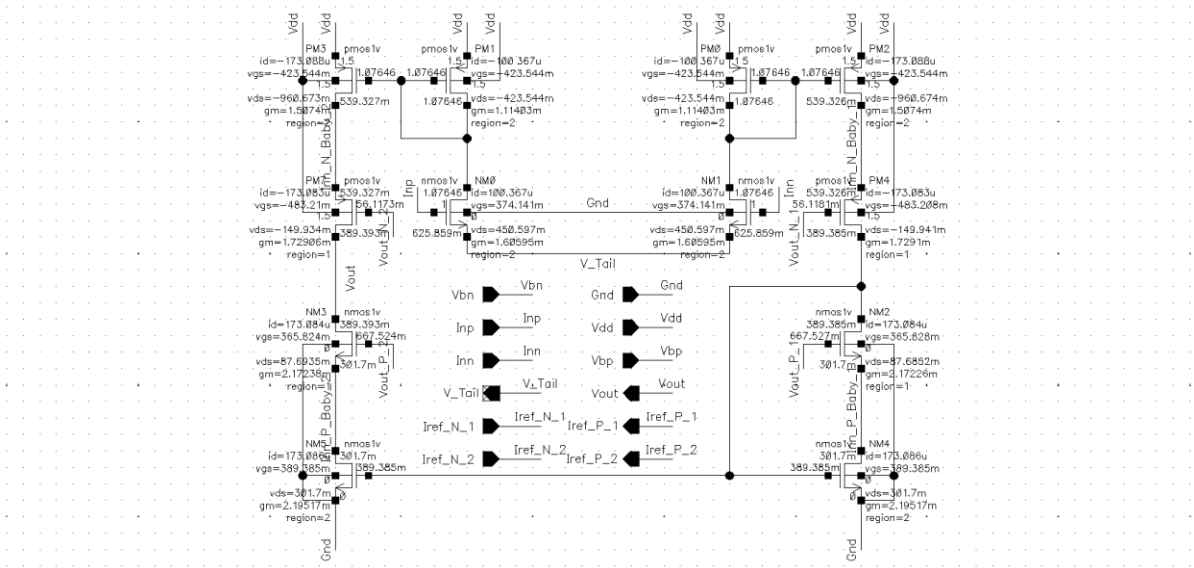
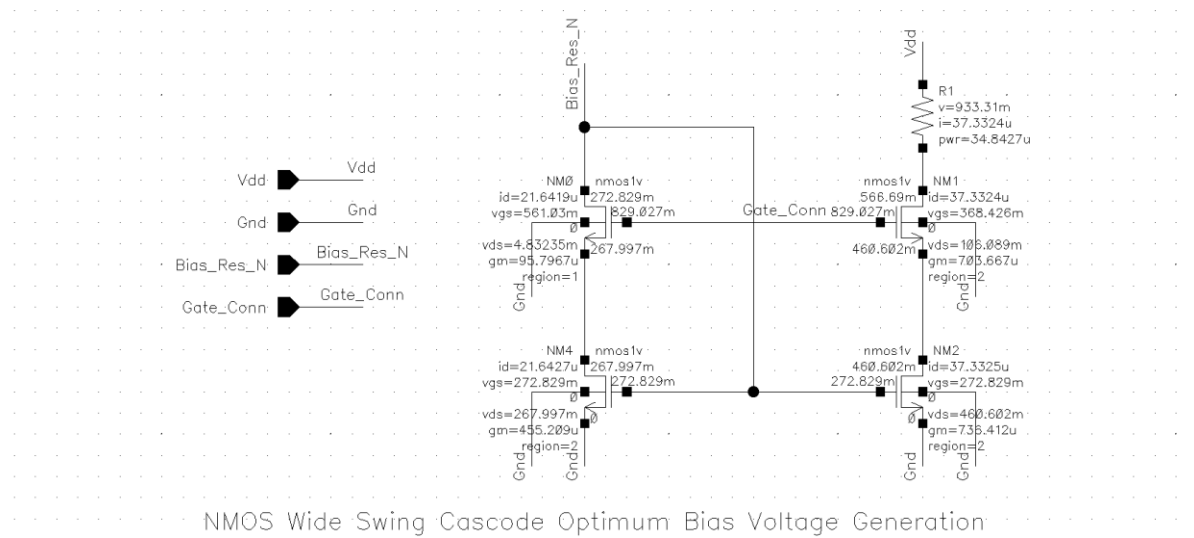
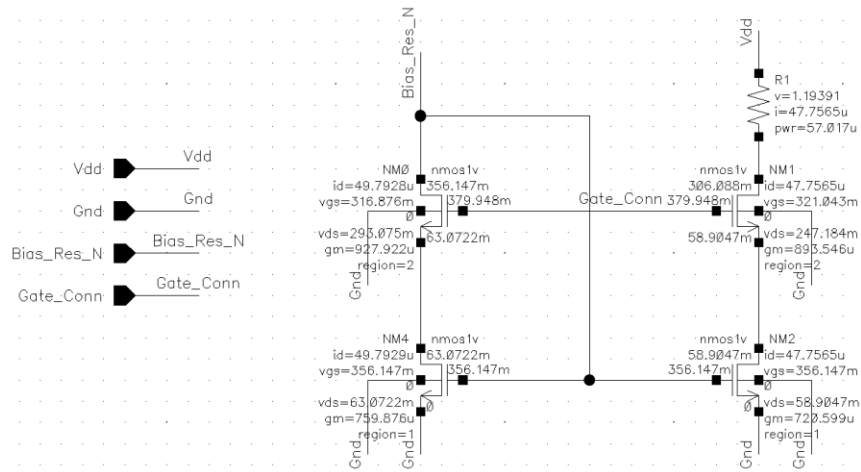


Figure 38: Current Mirror OTA Circuit



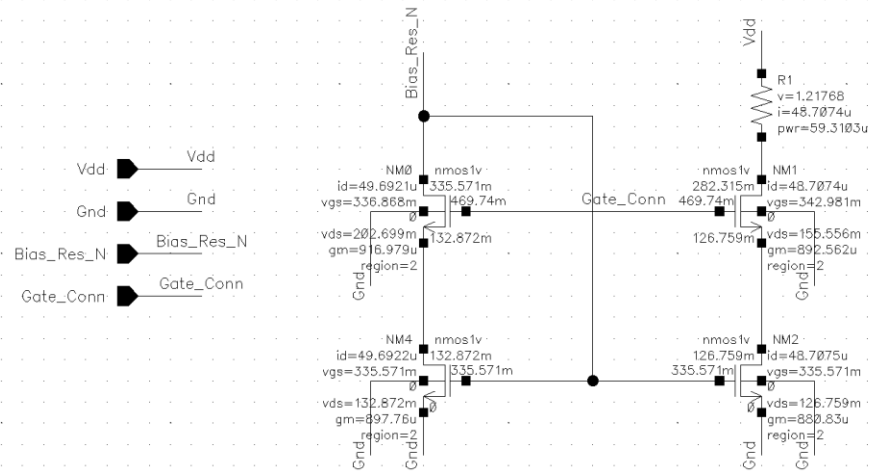
NMOS Wide Swing Cascode Optimum Bias Voltage Generation

Figure 39: NMOS Wide Swing Cascode Biasing Circuit



PMOS Wide Swing Cascode Optimum Bias Voltage Generation

Figure 40: PMOS Wide Swing Cascode Biasing Circuit



Tail Current Wide Swing Cascode Optimum Bias Voltage Generation

Figure 41: Tail Current Wide Swing Cascode Biasing Circuit

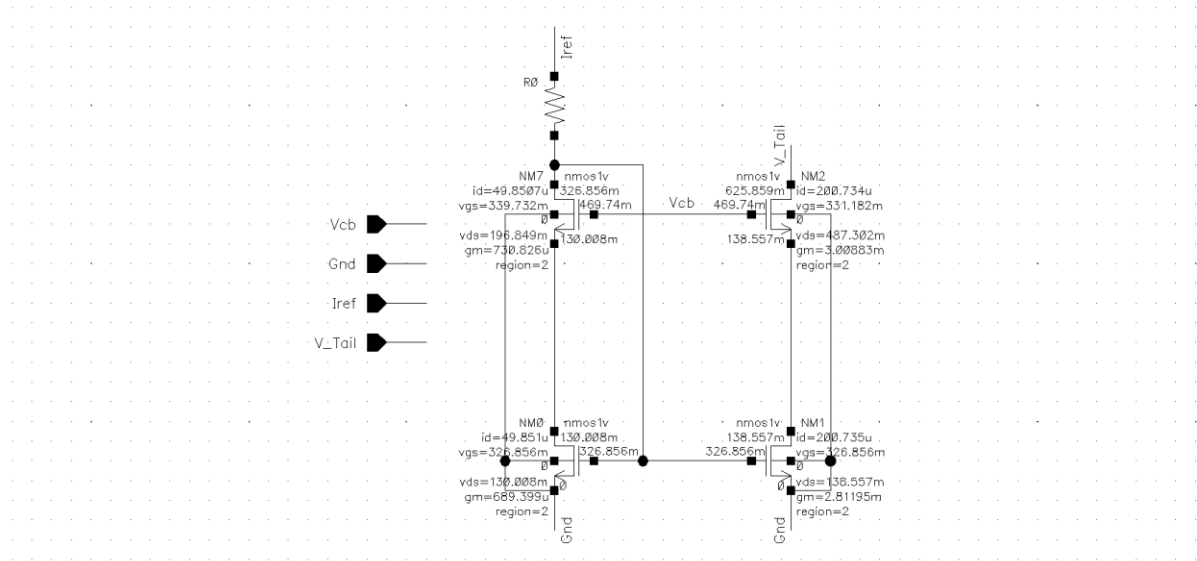


Figure 42: Tail Current Circuit

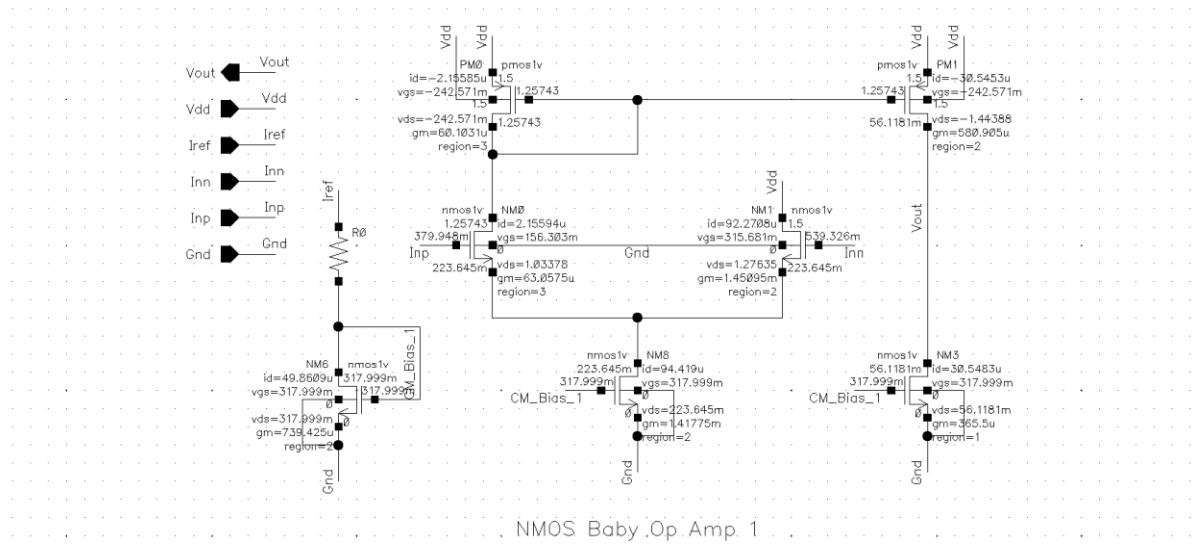


Figure 43: NMOS Baby Op Amp 1

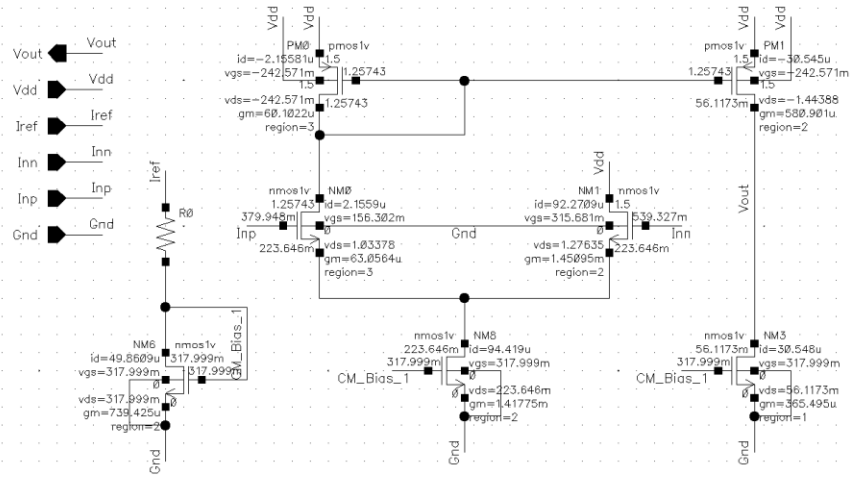


Figure 44: NMOS Baby Op Amp 2

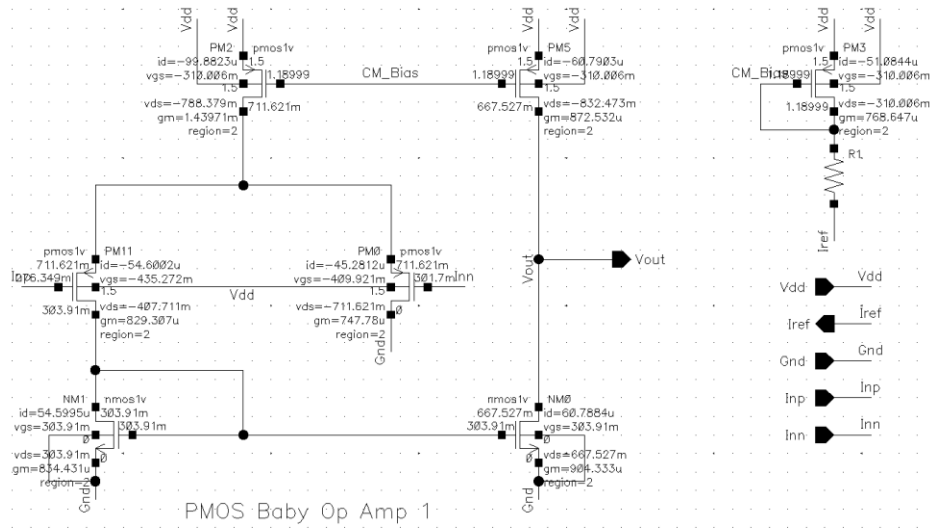


Figure 45: PMOS Baby Op Amp 1

Conclusion

In this project, an OTA using CMOS 90nm technology has been successfully designed, meeting 10 out of 12 specifications. However, these have been met in a specific condition: no output buffer. In case when the buffer is added, a reduced performance is observed, emphasizing the fact that design of a good buffer that consumes low power and does not hamper the gain is a tedious task. The designed OTA works for voltage range of 1.5V to 1.7V consuming 3mW of power. 68dB of gain is achieved with GBW of 409 MHz and 81 deg phase margin. Output swing of 0.93V was achieved and a wide input common mode range allows 0.93V overlap with output swing. The designed circuit has 109dB of CMRR and 93dB of PSRR which suggest excellent rejection to both common mode and power supply noise. The settling time is 16ns and the slew rate is 154 V/us which suggests excellent transient performance and fast response time of the circuit. To enhance the performance of the design it is recommended to change the operating current to a lower value that would allow for smaller devices with the same V_{eff} and higher output impedances to get higher gains. Another suggestion to enhance circuit performance is to design it for higher gain values to compensate for the loss by output buffer.

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